



Queues under Stochastic Priority Switching

Replication Results, Diagnostics, and Extensions

Stochastic Operations Research — Course Project

June 2026

Original study: Palmer, Panayidis, Knight, and Williams, JORS.

Executive Assessment

Experiment	Status	Evidence from current rerun
Overtaking distribution, Fig. 5	Partial	Workflow reproduced, but the paper candidate (3, 1) is not the best point in this run.
Stability / ADF check	Partial	Notebook rates reproduce the stable/unstable split; the paper-text stable case remains inconclusive.
Bounded Markov approximation	Supported	Both parameter variants converge by $b = 22$, with boundary quantities below 10^{-4} .
Effect of switching matrix H	Partial	Scenarios B and C pass the boundary threshold; Scenario A violates $Q(16) < 0.012$.
Extensions	Supported	Fine-grid calibration and sojourn-time variance are useful exploratory analyses.

Bottom line

The reproduction pipeline is complete and informative, but the numerical evidence should be reported as a mixed replication rather than a full success.

Replication Architecture

Inputs

- Original source code under `archive/original/src`
- Surgery waiting-list data from the original repository
- Paper-text parameters
- Original-notebook parameter variants

Reproduction layer

- Shared data, simulation, and metric helpers
- One script per paper experiment
- Parameter-variant runner for paper-text and notebook runs
- CSV metrics plus PNG figures for auditability

Main result folders: `results/parameter_variants/common`, `paper_text`, and `original_notebook`.

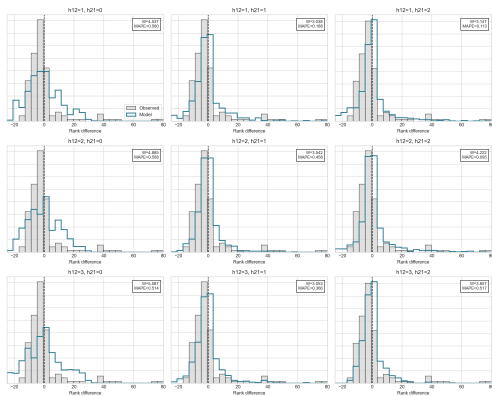
Why Two Parameter Variants Were Needed

Experiment	Paper-text setting	Notebook setting	Impact
Stability	$\theta_{12} = 3, \theta_{21} = 1$	$\theta_{12} = 2, \theta_{21} = 1$	Notebook setting reproduces ADF split; paper-text setting does not.
Bounded Markov / validation / variance	$\mu_2 = 2.5$	$\mu_2 = 1.6667$	Both converge, but produce different L and W levels.
Figure 5 and effect of H	Common run	Common run	No known paper-vs-notebook conflict in the current scripts.

Reporting rule

Use the paper-defined metrics for reproduction claims. Use composite calibration loss only in the extension section.

Figure 5: Overtaking Distribution

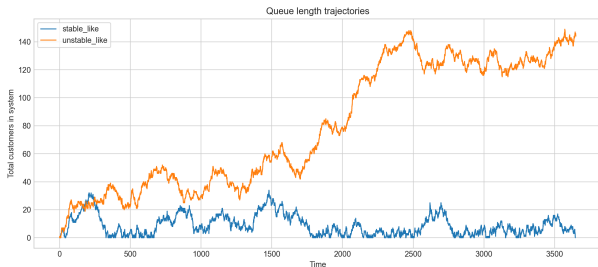


Criterion	Best point	Value
Wasserstein	(1, 1)	3.038
MAPE	(2, 2)	0.0946
Paper candidate	(3, 1)	W=3.053, MAPE=0.366

Interpretation

The paper candidate is close under Wasserstein distance, but not under waiting-time MAPE. The original Fig. 5 conclusion is therefore not fully reproduced by the current run.

Stability: ADF Diagnostics

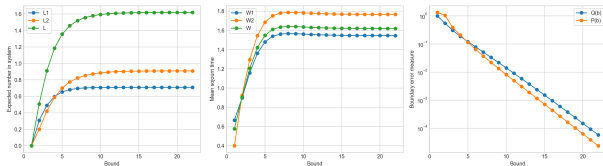


Variant	Stable case p	Unstable case p
Paper text	0.156	0.856
Notebook	0.00385	0.864

Interpretation

The stable/nonstationary contrast is reproduced under the original-notebook switching rates. The paper-text stable case remains visually bounded but fails the $p < 0.05$ ADF criterion.

Bounded Markov Approximation



Variant, $b = 22$	L	W	$Q(b)$
Paper text	1.620	1.620	5.88×10^{-5}
Notebook	1.916	1.916	7.61×10^{-5}

Interpretation

This is the strongest replication result. Truncation error becomes small in both parameter variants, and L and W stabilize as the bound grows.

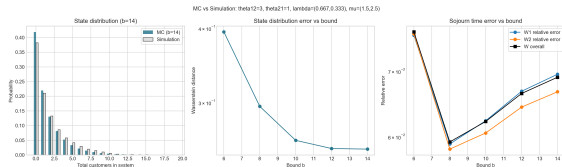
Effect of Switching Matrix H

Scenario	$Q(16)$	$P(16)$	$U_{\min}$	$U_{\max}$	Assessment
A: prioritized slower	0.0822	0.0120	1.704	7.927	Partial: P is within threshold, but Q exceeds 0.012 at 181 grid points.
B: equal rates	0.00753	0.00114	1.022	1.966	Supported: boundary checks pass across the grid.
C: prioritized faster	0.0102	0.00152	0.521	2.257	Supported: boundary checks pass across the grid.

Interpretation

The qualitative heatmap workflow is reproduced, but Scenario A should not be described as satisfying the paper's stated boundary check.

Markov Chain vs Simulation

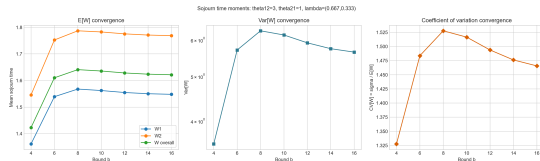


Variant	W-error range	State-W range
Paper text	5.93–7.68%	0.251–0.395
Notebook	4.43–6.29%	0.104–0.308

Interpretation

The validation is directionally reasonable, especially for state distributions, but the simulation horizon run is short relative to the paper-scale long run.

Sojourn-Time Variance

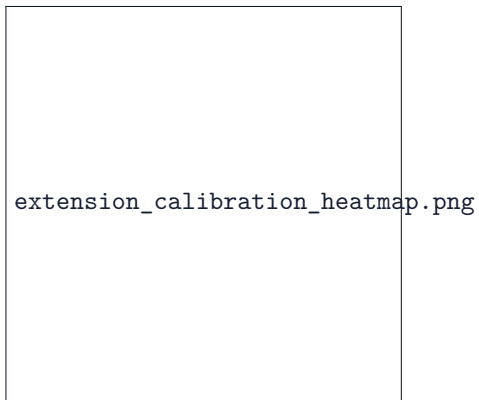


Variant	b	W	$\text{Var}(W)$
Paper text	16	1.622	5.649
Notebook	16	1.918	7.231

Interpretation

The second-moment computation is internally consistent and useful. It should be presented as part reproduction of the absorbing-chain moment formulas plus additional visualization.

Extension: Fine-Grid Calibration



Criterion	Point	W	MAPE
Best calibration loss	(3, 2)	2.657	0.179
Best Wasserstein	(1, 2)	2.576	0.285
Best MAPE	(3.5, 2)	3.368	0.108
Paper point	(3, 1)	3.977	0.518

Interpretation

The composite loss is useful for screening the calibration landscape, but it is not a paper metric and is not normalized. It supports sensitivity analysis, not a definitive claim.

Main Takeaways

What worked

- The repository now produces a complete, auditable result set for all implemented scripts.
- Bounded Markov convergence is robust under both parameter variants.
- The original-notebook stability experiment reproduces the expected split.
- The extension analyses are reasonable when clearly labeled as exploratory.

What needs caution

- Fig. 5 does not confirm $(\theta_{12}, \theta_{21}) = (3, 1)$ as the unique or best fit.
- Paper-text stability and Scenario A effect-of- H boundary checks are not fully reproduced.
- MC-vs-sim validation needs longer simulation and confidence intervals before stronger claims.



Thank you

Questions and discussion